The Goods Market

Intermediate Macroeconomics

Fall 2025, Sciences Po. Paris Campus du Havre

Lecture 7

Announcements

- Textbook: Chapter 3
- **Take Home Midterm will be published on Thursday (10/17)** on Moodle, and you will have one week to solve that exam


DATA SHOWS YOU CONSUMERS AREN'T SPENDING ENOUGH.
OPEN YOUR WALLETS! BUY SOME BIG TICKET ITEMS.
WHEN OUR SALES START TO PICK UP...
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MAYBE. AT HALF — YOUR OLD PAY.
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3. Changes in **income** lead to changes in the **demand for goods**

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4. **Exports (X)**: purchases of U.S. goods and services by foreigners
5. **Imports (IM)**: purchases of foreign goods and services by U.S. consumers, U.S. firms and the U.S. government

Trade balance

$$\text{Trade Balance} = X - IM$$

- export $>$ import $\implies$ trade surplus
- export $<$ import $\implies$ trade deficit

The Demand for Goods

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- This is a behavioral equation that captures the behavior of consumers.

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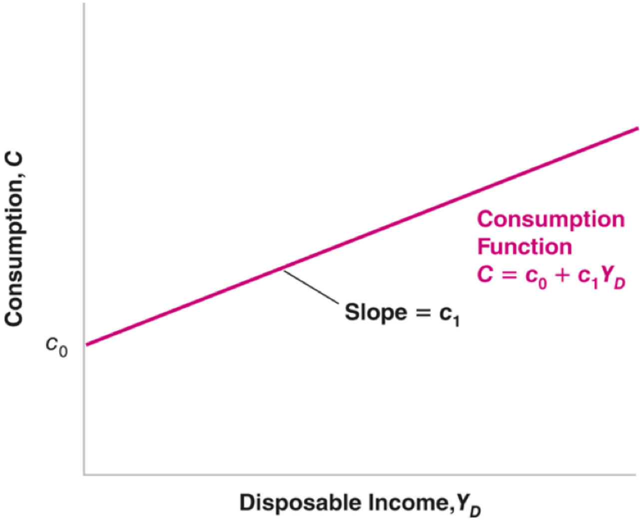
  - $c_0 > 0$
  - How? "dissave": borrowing, or selling assets.

Consumption



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We start by taking investment as given (exogenous): $\bar{I}$

$$I = \bar{I}$$

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We will think about G and T as exogenous variables: governments do not behave with the same regularity as consumer or firms.

Equilibrium Output

Equilibrium Output

$$Z \equiv C + I + G$$
$$Z \equiv c_0 + c_1(Y - T) + \bar{I} + G$$

Equilibrium in the goods market requires that:
production (Y) = demand (Z)

$$Y = c_0 + c_1(Y - T) + \bar{I} + G$$

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Autonomous spending

- Part of demand for goods that **does not depend on output** $\Longrightarrow$ is exogenous
- If government is running a balanced budget ($T = G$) and $0 < c_1 < 1$, then $G - c_1 T$ is positive
- We will always assume that the autonomous spending is positive

Equilibrium Output: Graph

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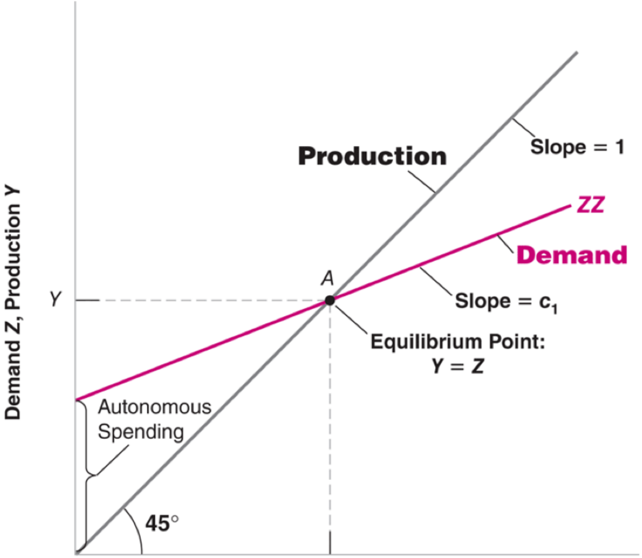
- In equilibrium, **production equals demand**

Graph: Keynesian cross

Equilibrium: **production is equal to demand**.



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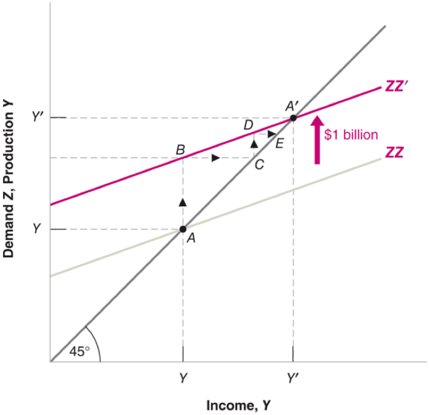
Example: increase in c_0



Equilibrium Output: Algebra

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- Multiplier's size depends on the propensity to consume (c_1)

Multiplier: Example

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- An increase of autonomous household consumption by \$1 billion will increase output by $2.5 \times$ \$1 billion $=$ \$2.5 billion.

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- Intuition behind the multiplier's effect: When income is spent, this spending becomes someone else's income, and so on ...

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This is the multiplier!

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- We would get the same result if we start with an increase of Investment ($\bar{I}$)
- We would also get the same result if we start with an increase of Government spending ($\bar{G}$)

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  - For taxes, the multiplier effect will be:

$$\frac{-c_1}{1 - c_1}$$

IS Relationship

Investment $=$ Saving

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$$\text{Saving} = \text{Private Saving} + \text{Public Saving}$$
$$\text{Saving} = S + (T - G)$$

Investment = Saving

Private Saving: saving by consumers (households)

$$S = Y_D - C$$
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Public Saving: saving by the government

$$T - G$$

- if $T - G > 0$, Budget surplus (government actually saves)
- if $T - G < 0$, Budget deficit (government borrows)

Investment = Saving

Let's look at the equilibrium in the goods market:

$$Y = C + I + G$$

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We often simplify this by assuming that the government runs a balanced budget ($G = T$)

$$I = S$$

Paradox of saving

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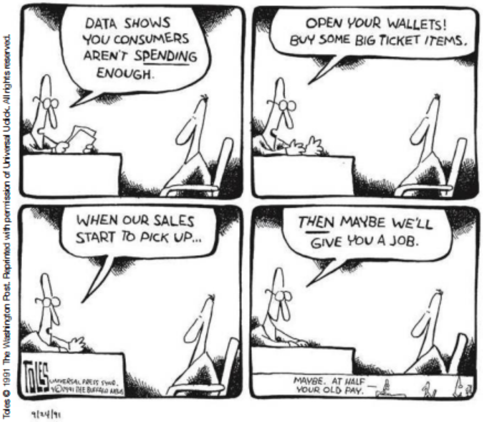
We are often told about the virtues of saving but this model tells us a different story

Recall the first slide?



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- Increase in initial saving generates lower equilibrium output

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- As people attempt to save more, the result is both an output decline and unchanged saving
- This surprising result is known as the paradox of saving

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Real-life example: Austerity measures in Europe (e.g. Greece)